



Winning Space Race with Data Science

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Correct Partial Needs Work

[+] CORRECT

criterion_0

[CORRECT] **Question 1: Did the learner uploaded the URL of GitHub repository, including all the completed notebooks and Python files Note:Note: Evaluate the submitted PDF to ensure it includes completed slides with relevant content for the following sections: Executive Summary, Introduction, Data Collection and Data Wrangling Methodology, EDA and Data Visualization Methodology, Predictive Analysis (Classification), EDA with Visualization, EDA with SQL Slides/Queries, Folium Map, Plotly Dash-related content, Predictive Analysis Results, and Conclusion. Slides must contain meaningful explanations, analysis, visuals, or results. Award 0 points if the slide contains only a heading, title, or keywords without supporting content. Question 2: Did the learner upload the final presentation as a .pdf file? Question 3: Did the learner complete the required Executive Summary slide? Question 4: Did the learner complete the required Introduction slide? Question 5: Did the learner complete the required data collection and data wrangling methodology slides? Question 6: Did the learner complete the required EDA and data visualization methodology slides? Question 7: Did the learner complete the predictive analysis (classification) slide? Question 8: Did the learner complete the required EDA with visualization slides? Question 9: Did the learner complete the required EDA with SQL slides/queries? Question 10: Did the learner complete the required Folium map slides? Question 11: Did the learner complete the required Plotly Dash-related slides? Question 12: Did the learner complete the predictive analysis results slides based on the following criteria? Question 13: Did the learner complete the Conclusion slide? The completed Conclusion slide should include at least: Question 14: Did the learner apply any creativity to this presentation? Creativity may

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data was collected via the SpaceX REST API and by web scraping historical Falcon 9 launch records from Wikipedia, then cleaned and labeled for landing outcome
- Exploratory analysis (visualization + SQL) and interactive analytics (Folium map, Plotly Dash dashboard) were used to uncover patterns between payload, orbit, launch site and landing success
- Four classification models (Logistic Regression, SVM, Decision Tree, KNN) were trained and tuned with GridSearchCV; the Decision Tree Classifier achieved the best cross-validated accuracy (87.5%)
- Findings show landing success has improved substantially over time and varies meaningfully by launch site, orbit type and payload mass
- GitHub link: [leonorrebola9/datascience_capstone](https://github.com/leonorrebola9/datascience_capstone)

Introduction

- SpaceX advertises Falcon 9 launches at \$62 million, versus \$165+ million from other providers, largely because SpaceX reuses the rocket's first stage
- If we can predict whether the first stage will successfully land, we can estimate the cost of a launch — valuable for a competitor bidding against SpaceX
- This project aims to answer: what factors most influence a successful first-stage landing, and can we build a reliable model to predict it?

Section 1

Methodology

Methodology

Executive Summary

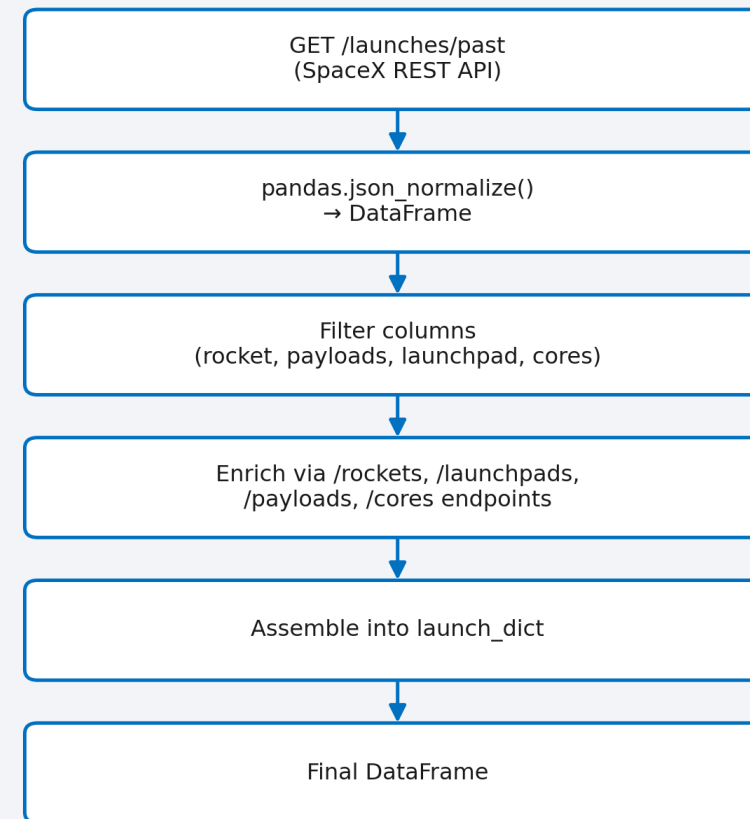
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Two data sources were combined: the SpaceX REST API (structured launch/rocket/payload/core data) and Wikipedia (historical Falcon 9 and Falcon Heavy launch records via web scraping)
- API data was requested with the Python requests library and normalized into a DataFrame with `pandas.json_normalize()`
- Wikipedia data was parsed with BeautifulSoup, extracting the launch records table and cleaning inconsistent HTML annotations

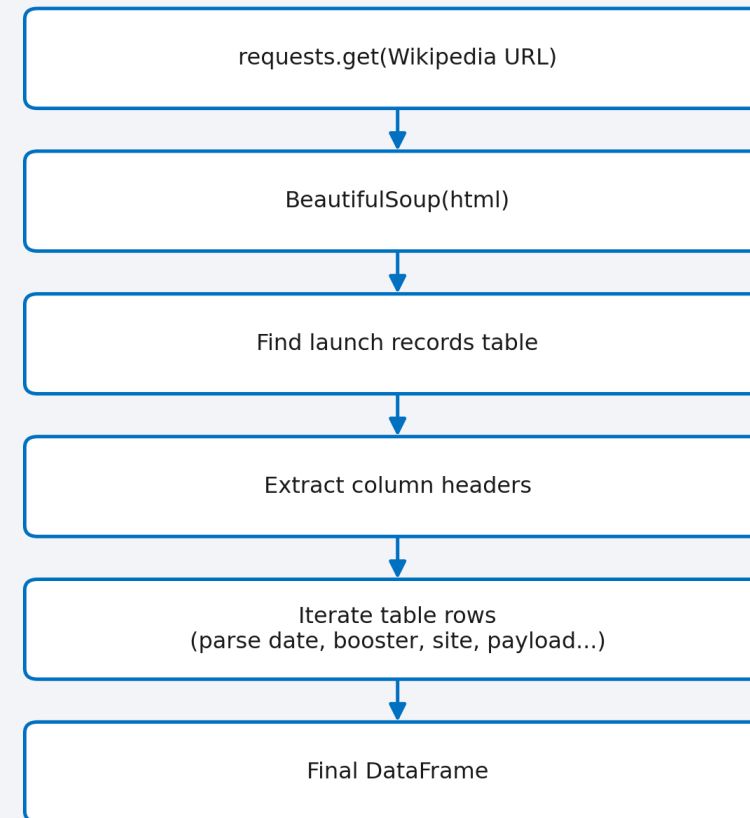
Data Collection – SpaceX API

- Process: GET request to the SpaceX API, then enrich and assemble into a DataFrame (see diagram)
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/jupyter-labs-spacex-data-collection-api-v2.ipynb



Data Collection - Scraping

- Process: request the Wikipedia page, parse the HTML table with BeautifulSoup, and build a DataFrame (see diagram)
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/jupyter-labs-webscraping.ipynb



Data Wrangling

- Missing values were identified (`df.isnull().sum()/len(df)*100`); PayloadMass NaNs were replaced with the column mean, while LandingPad NaNs were kept as None (meaning no landing pad was used)
- Launch counts were summarized per site (CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13) and per orbit type (GTO, ISS, VLEO, PO, LEO, SSO, MEO, HEO, ES-L1, SO, GEO)
- The Outcome column (e.g. 'True ASDS', 'False Ocean') was converted into a binary Class label: 0 for an unsuccessful landing, 1 for a successful one
- Overall landing success rate across the cleaned dataset: 66.7%
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/labs-jupyter-spacex-Data%20wrangling-v2.ipynb

EDA with Data Visualization

- Scatter plots (Flight Number vs. Launch Site, Payload vs. Launch Site, Flight Number vs. Orbit Type, Payload vs. Orbit Type) were used to reveal how launch success (color-coded) relates to site, payload mass and orbit over time
- A bar chart of success rate by orbit type highlighted which orbits are most reliably associated with successful landings
- A line chart of yearly average success rate showed the overall improvement trend in landing success as SpaceX's technology matured
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

- Loaded the SpaceX.csv dataset into a SQLite database table (SPACEXTABLE) via `pandas.to_sql()`, then queried it with SQL magic (`%sql`)
- Queries covered: unique launch sites, sample records, total/average payload mass by customer and booster version, first successful ground-pad landing date, boosters with drone-ship success in a payload range, mission outcome counts, booster(s) with maximum payload, 2015 drone-ship failures by month, and a ranked count of landing outcomes over a date range
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- `folium.Circle` and `folium.Marker` were added for each of the 4 launch sites, using their latitude/longitude, to visualize their geographic spread and proximity to the equator/coast
- A `MarkerCluster` grouped individual launch records, color-coded green (success) or red (failure), so success patterns per site are visible at a glance
- `MousePosition`, a `PolyLine` and a distance marker were added to measure and display the distance from a launch site to its nearest coastline, highlighting SpaceX's preference for coastal/low-population launch sites
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

- A dropdown lets the user pick 'All Sites' or a single launch site, driving a pie chart of success counts (or success vs. failure split for one site)
- A range slider (0-10,000 kg) filters a scatter plot of Payload Mass vs. launch outcome (class), color-coded by Booster Version Category, updating interactively as the range changes
- These interactions let a user quickly compare success rates across sites and see how payload mass and booster version relate to landing success
- https://github.com/leonorrebola9/datascience_capstone/blob/main/dashboard/spacex-dash-app.py

Predictive Analysis (Classification)

- Features were standardized with StandardScaler, then split into train/test sets (80/20, random_state=2)
- Four classifiers — Logistic Regression, SVM, Decision Tree, KNN — were each tuned via GridSearchCV (10-fold cross-validation) over a model-specific hyperparameter grid
- Each tuned model's cross-validated accuracy and test-set accuracy were compared, along with its confusion matrix, to select the best performer
- The Decision Tree Classifier had the highest cross-validated accuracy (87.5%), making it the recommended model
- https://github.com/leonorrebola9/datascience_capstone/blob/main/notebooks/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

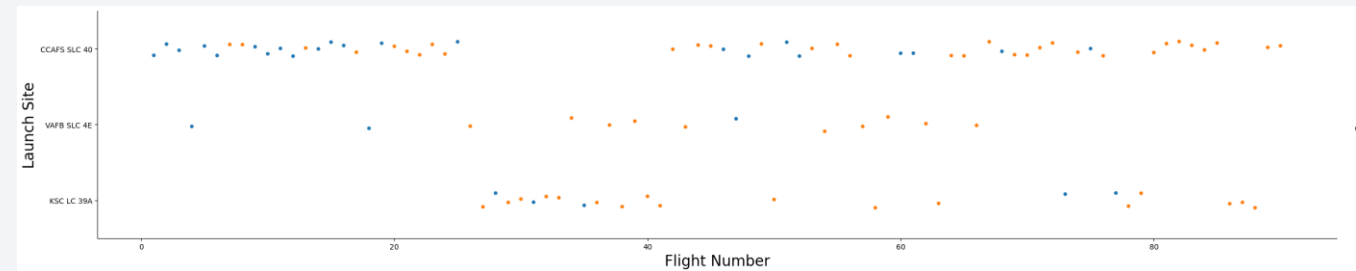


Section 2

Insights drawn from EDA

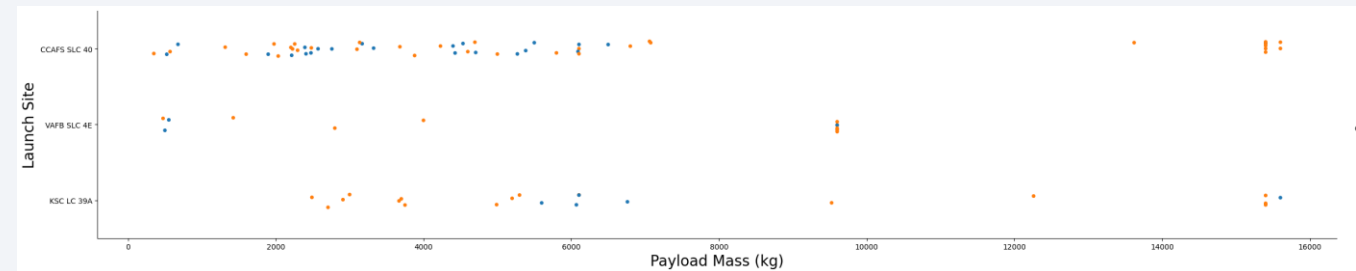
Flight Number vs. Launch Site

- Chart: seaborn catplot, Launch Site (y) vs Flight Number (x), colored by outcome (Class 0=blue=did not land, Class 1=orange=landed)
- CCAFS SLC-40 was used from flight #1 and shows a mix of outcomes throughout, with orange (success) becoming denser at higher flight numbers
- KSC LC-39A only came online around flight #28 onward — later in the program — and already shows a healthy share of successful (orange) landings



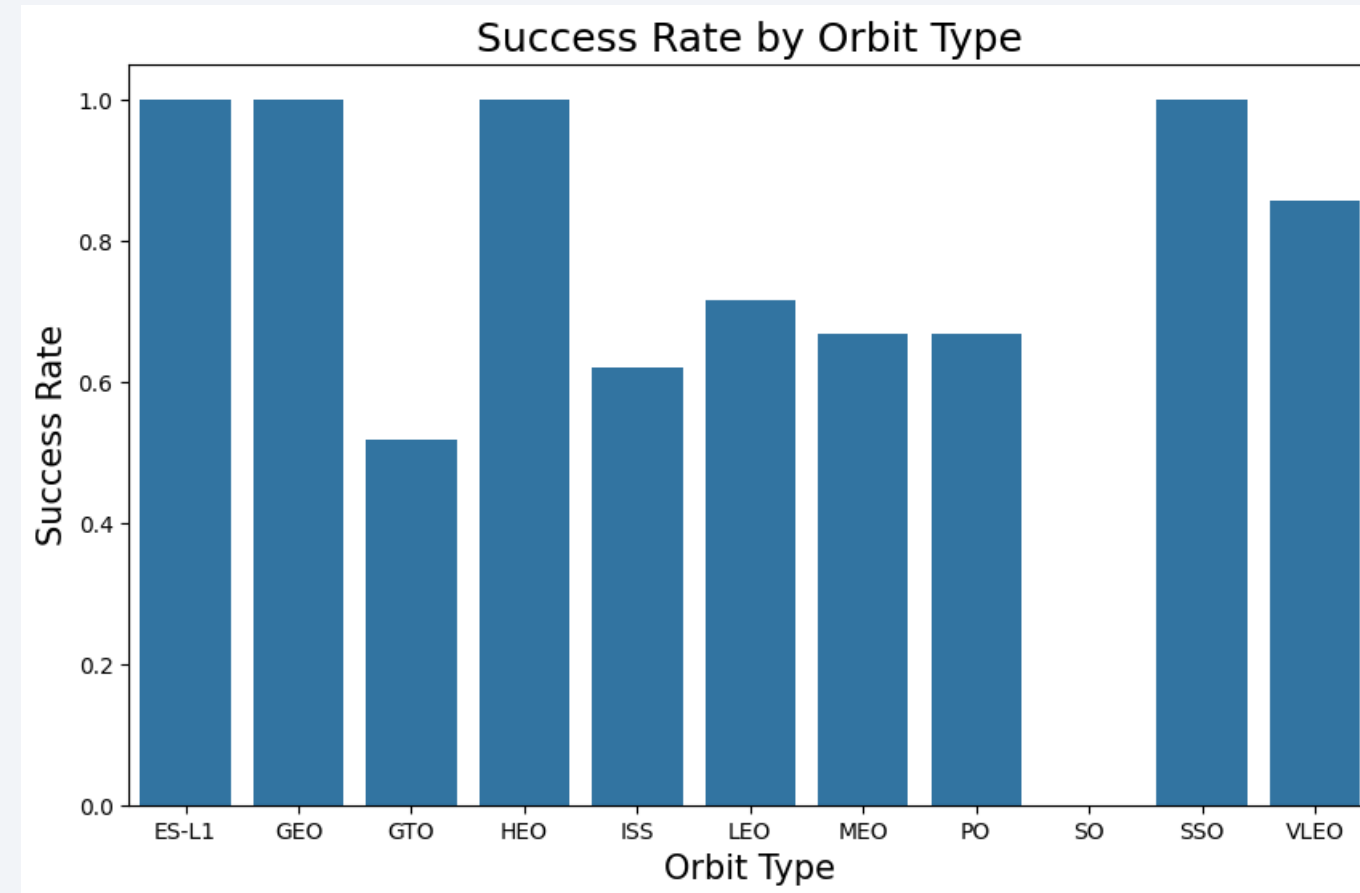
Payload vs. Launch Site

- Chart: seaborn catplot, Launch Site (y) vs Payload Mass (x), colored by landing outcome
- VAFB SLC-4E never launched payloads heavier than ~10,000 kg in this dataset
- CCAFS SLC-40 and KSC LC-39A both handle a wide range of payload masses, including the heaviest



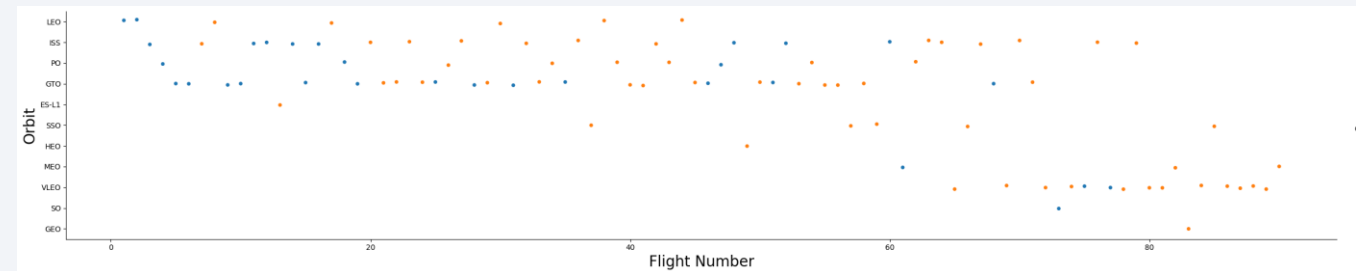
Success Rate vs. Orbit Type

- Chart: bar chart of mean Class (success rate) grouped by Orbit type
- ES-L1, GEO, HEO and SSO orbits show a 100% landing success rate in this dataset (though based on few launches)
- GTO has the lowest success rate (~52%); VLEO is strong (~86%) reflecting many recent, mature-era launches



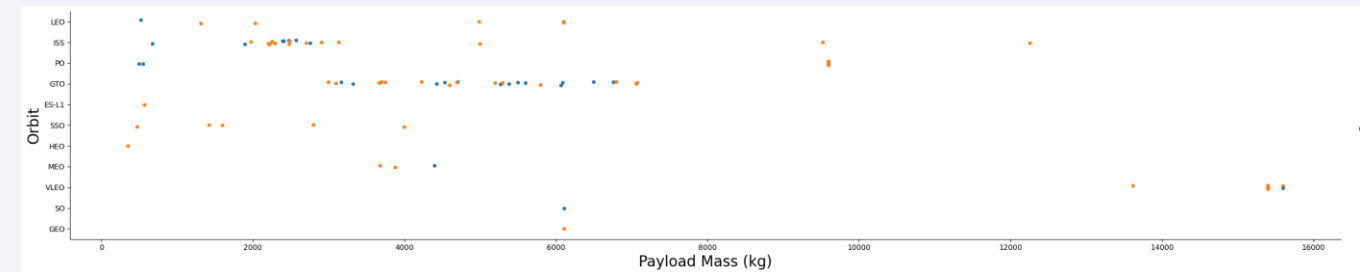
Flight Number vs. Orbit Type

- Chart: seaborn catplot, Orbit (y) vs Flight Number (x), colored by landing outcome
- In LEO, success is clearly related to flight number — later flights (higher numbers) succeed far more often
- GTO shows failures spread across both early and later flight numbers, a harder orbit to land from consistently



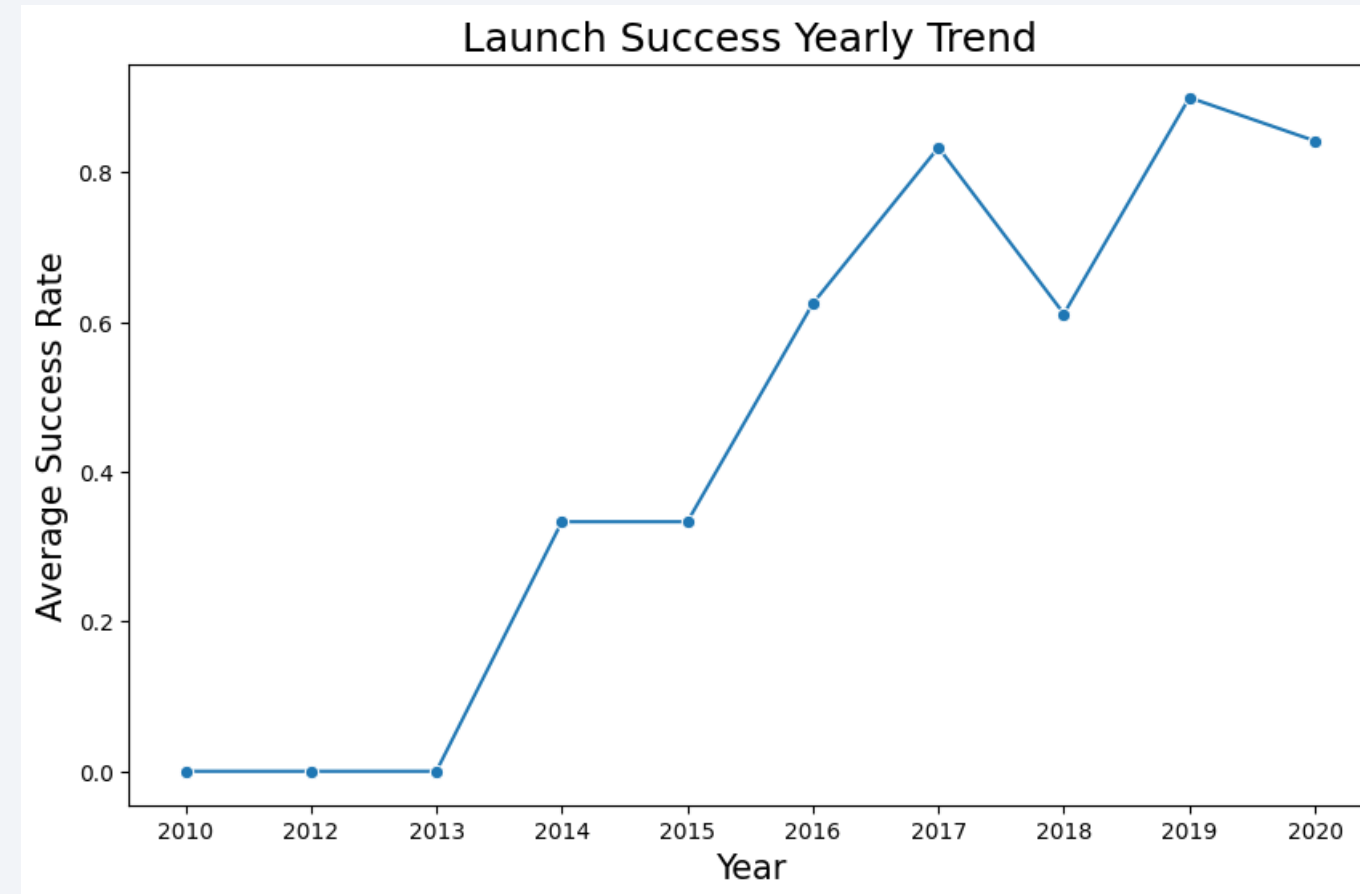
Payload vs. Orbit Type

- Chart: seaborn catplot, Orbit (y) vs Payload Mass (x), colored by landing outcome
- Heavier payloads to Polar, LEO and ISS orbits tend to have a higher positive landing rate
- It's hard to distinguish a clear payload-success pattern for GTO, given its mix of outcomes



Launch Success Yearly Trend

- Chart: line chart of average landing success rate by year
- Success rate rose sharply from 2013 to 2017, then stayed roughly stable through 2018-2020 (with a dip in 2018)
- This trend reflects SpaceX's iterative booster improvements (v1.0 → v1.1 → FT → Block 5)



All Launch Site Names

- Query: `SELECT DISTINCT Launch_Site FROM SPACEXTABLE;`
- Result: 4 unique launch sites — CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- These are the only pads SpaceX has used for Falcon 9/Heavy launches in this dataset

Launch Site Names Begin with 'CCA'

- Query: `SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`
- Result: 5 earliest CCAFS LC-40 launches (2010-2013), including the first-ever Falcon 9 flight
- Confirms CCAFS LC-40 was SpaceX's original launch pad before other sites were added

Total Payload Mass

- Query: `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer='NASA (CRS)';`
- Result: 45,596 kg total payload mass carried for NASA's Commercial Resupply Services (CRS) missions
- Highlights NASA CRS as one of SpaceX's major early customers

Average Payload Mass by F9 v1.1

- Query: `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version='F9 v1.1';`
- Result: 2,928.4 kg average payload mass for the F9 v1.1 booster version
- F9 v1.1 was an early-generation booster with more limited payload capacity than later versions

First Successful Ground Landing Date

- Query: `SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome='Success (ground pad)';`
- Result: 2015-12-22 — the date of SpaceX's first successful ground-pad landing
- This was a historic milestone, proving controlled first-stage recovery was viable

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query: boosters with Landing_Outcome='Success (drone ship)' AND payload mass between 4000-6000 kg
- Result: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- All four are 'Full Thrust' (FT) boosters, showing this generation reliably handled mid-range payloads on drone-ship landings

Total Number of Successful and Failure Mission Outcomes

- Query: `SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;`
- Result: Success — 98, Success (payload status unclear) — 1, Success (with extra note) — 1, Failure (in flight) — 1
- Mission (payload delivery) outcomes are overwhelmingly successful — 100/101 missions succeeded

Boosters Carried Maximum Payload

- Query: booster_version with the maximum PAYLOAD_MASS__KG_ (via subquery with MAX())
- Result: 12 booster serials tied for the maximum payload mass, all Block 5 (F9 B5) boosters, e.g. B1048.4, B1049.4, B1051.3, B1056.4...
- Shows Block 5 is SpaceX's highest-payload-capacity booster generation

2015 Launch Records

- Query: month, Landing_Outcome, Booster_Version, Launch_Site WHERE Landing_Outcome='Failure (drone ship)' AND year=2015
- Result: January — F9 v1.1 B1012 at CCAFS LC-40; April — F9 v1.1 B1015 at CCAFS LC-40
- Both 2015 drone-ship failures involved the earlier F9 v1.1 booster, before landing technology matured

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

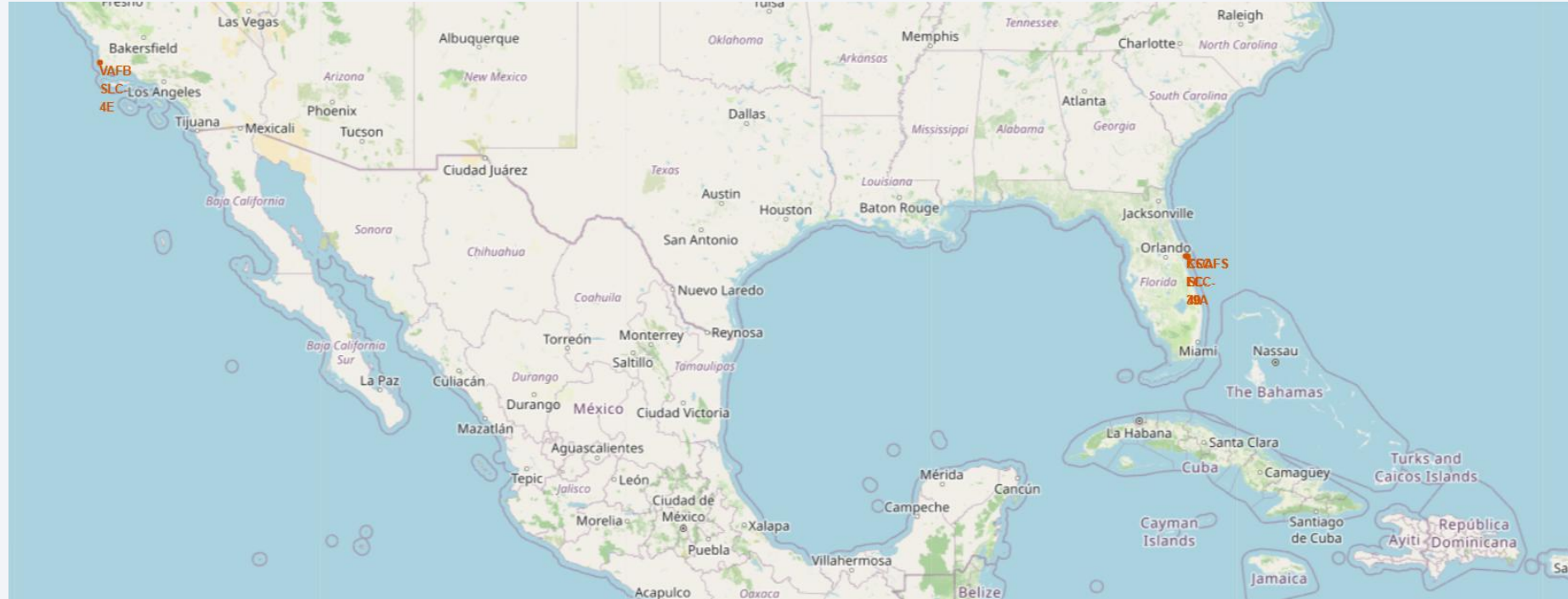
- Query: Landing_Outcome ranked by COUNT(*) between 2010-06-04 and 2017-03-20, descending
- Result (top to bottom): No attempt (10), Success (drone ship) (5), Failure (drone ship) (5), Success (ground pad) (3), Controlled (ocean) (3), Uncontrolled (ocean) (2), Failure (parachute) (2), Precluded (drone ship) (1)
- 'No attempt' still leads in this early period — landing attempts only became routine as the program matured

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a wide-angle shot from a high altitude, looking down at the Earth's surface. The horizon is visible as a thin line separating the dark blue of the sky from the lighter blue of the Earth's atmosphere. Below the horizon, the Earth's surface is covered in a dense network of city lights, appearing as bright yellow and orange spots. The lights are concentrated in certain areas, particularly along the coastlines and in the central part of the image. The overall color palette is dominated by deep blues and blacks, with the bright lights providing a stark contrast.

Section 3

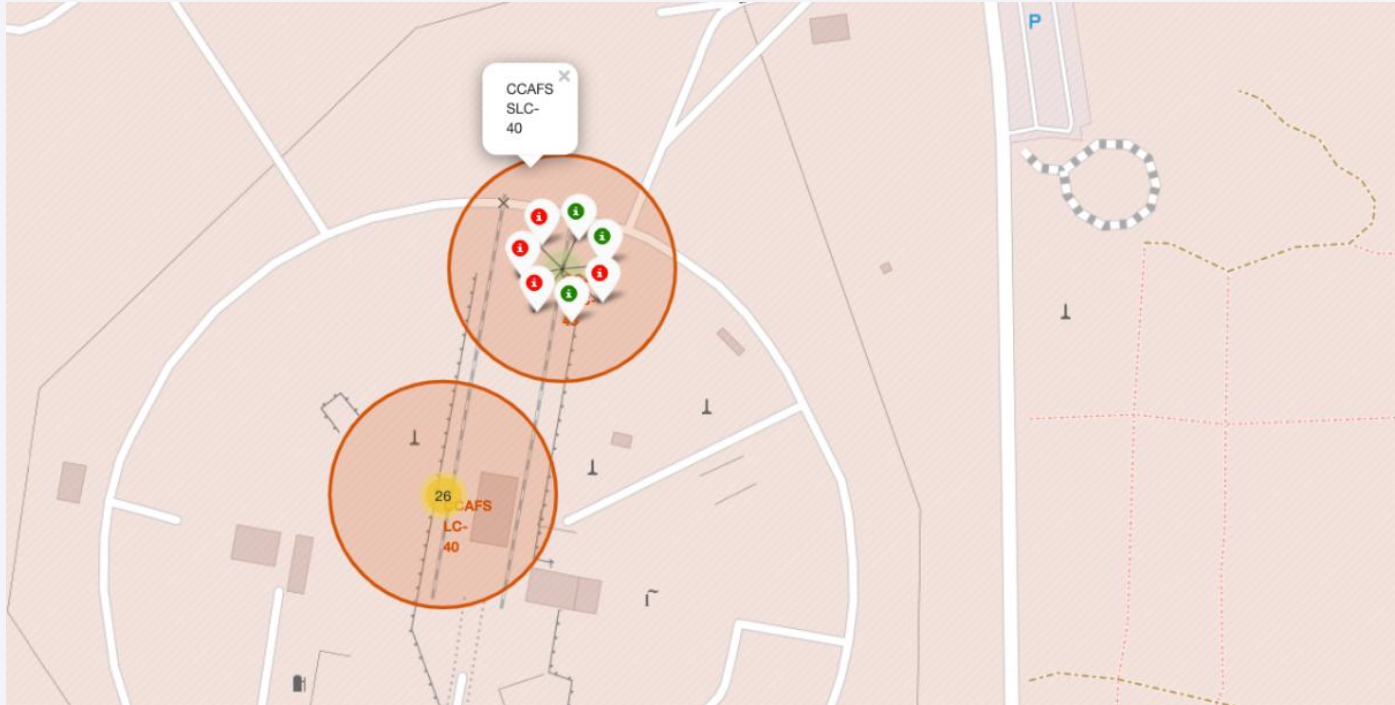
Launch Sites Proximities Analysis

All Launch Sites on a Global Map



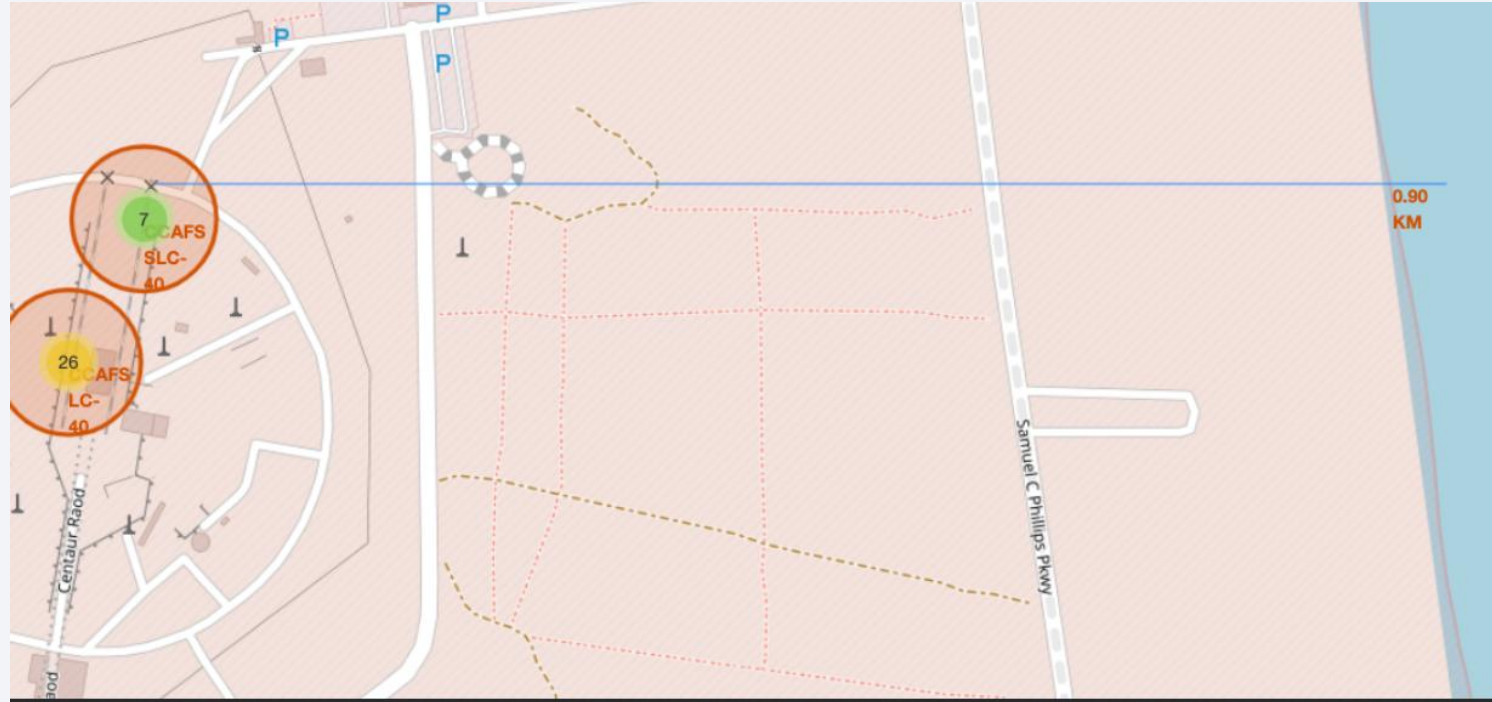
- Circle + text markers were added for all 4 Falcon 9 launch sites (CCAFS LC-40/SLC-40, KSC LC-39A, VAFB SLC-4E), overlaid on a map of the continental US
- All sites sit close to the coast (Florida east coast and California west coast), confirming SpaceX's preference for coastal, low-population launch corridors

Launch Outcomes by Site (Success/Failure Markers)



- Individual launch markers were color-coded green (success) and red (failure) and grouped with a MarkerCluster, so success patterns per site are visible when zoomed in
- Zooming into CCAFS shows a mix of green and red pins at SLC-40, illustrating both early failures and later successes at the same pad

Distance from Launch Site to Nearest Coastline



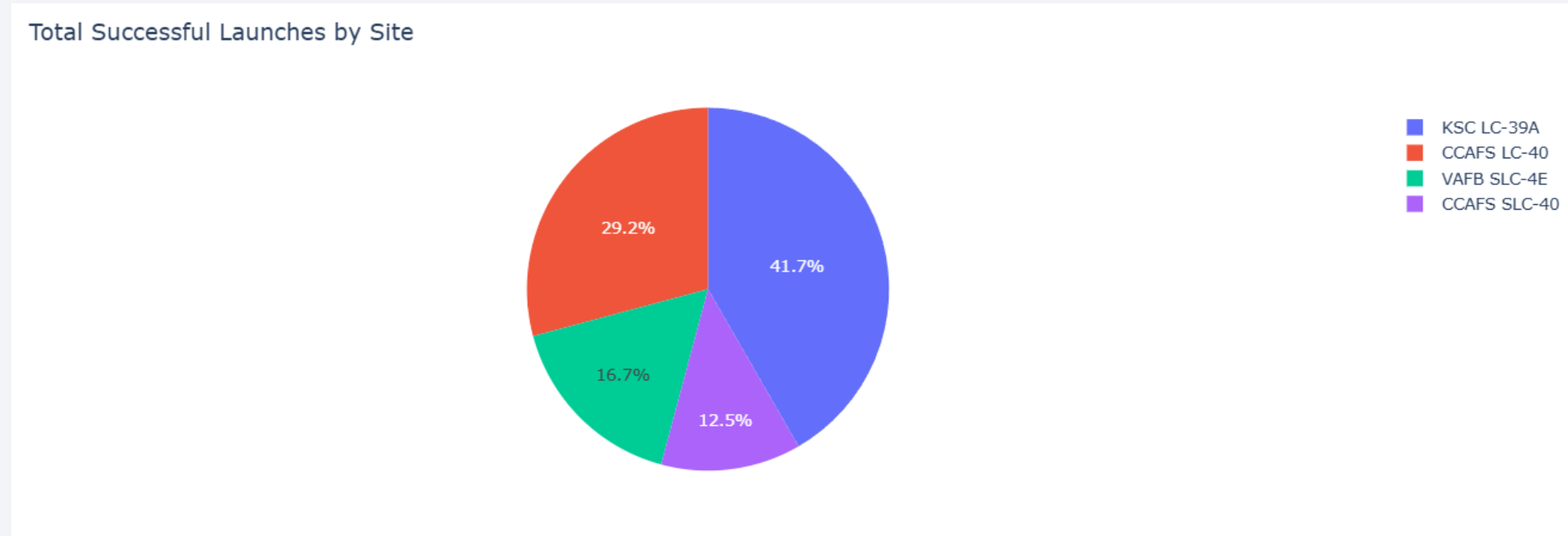
- MousePosition, a PolyLine and a distance marker were used to measure the distance from a launch pad to its nearest coastline point
- The measured distance here is 0.90 km, showing launch pads are sited extremely close to open water — consistent with range-safety practice for downrange rocket debris



Section 4

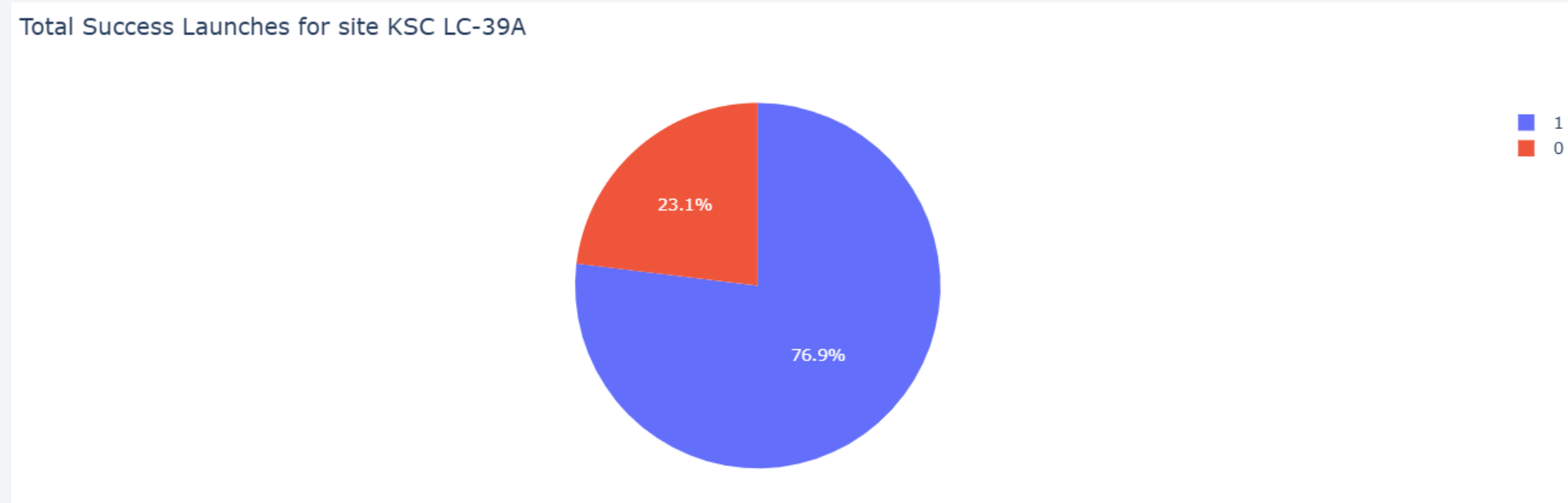
Build a Dashboard with Plotly Dash

Launch Success Count — All Sites



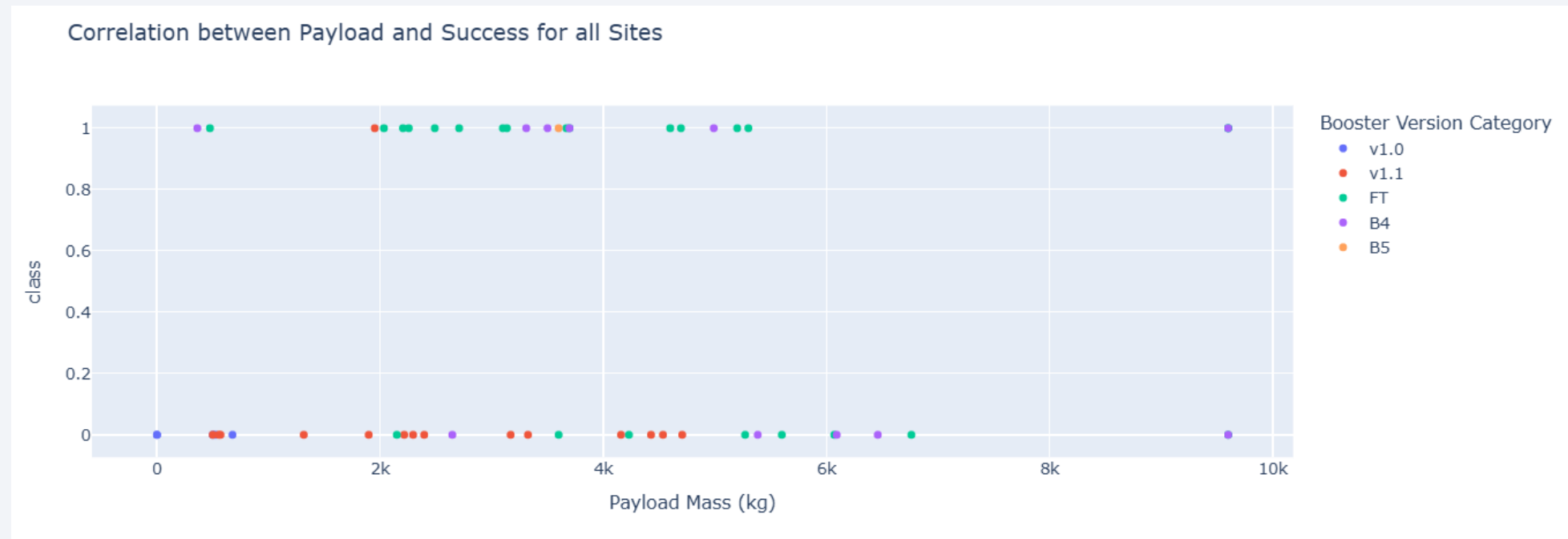
- With the dropdown set to 'All Sites', the pie chart shows the share of total successful launches contributed by each site
- KSC LC-39A contributes the most successful launches overall (41.7%), followed by CCAFS LC-40 (29.2%)

Highest Success-Rate Site — Pie Chart



- Selecting KSC LC-39A in the dropdown shows its individual success vs. failure split
- KSC LC-39A has the highest success rate of any site: 76.9% successful landings vs. 23.1% failures

Payload vs. Launch Outcome — Scatter Plot



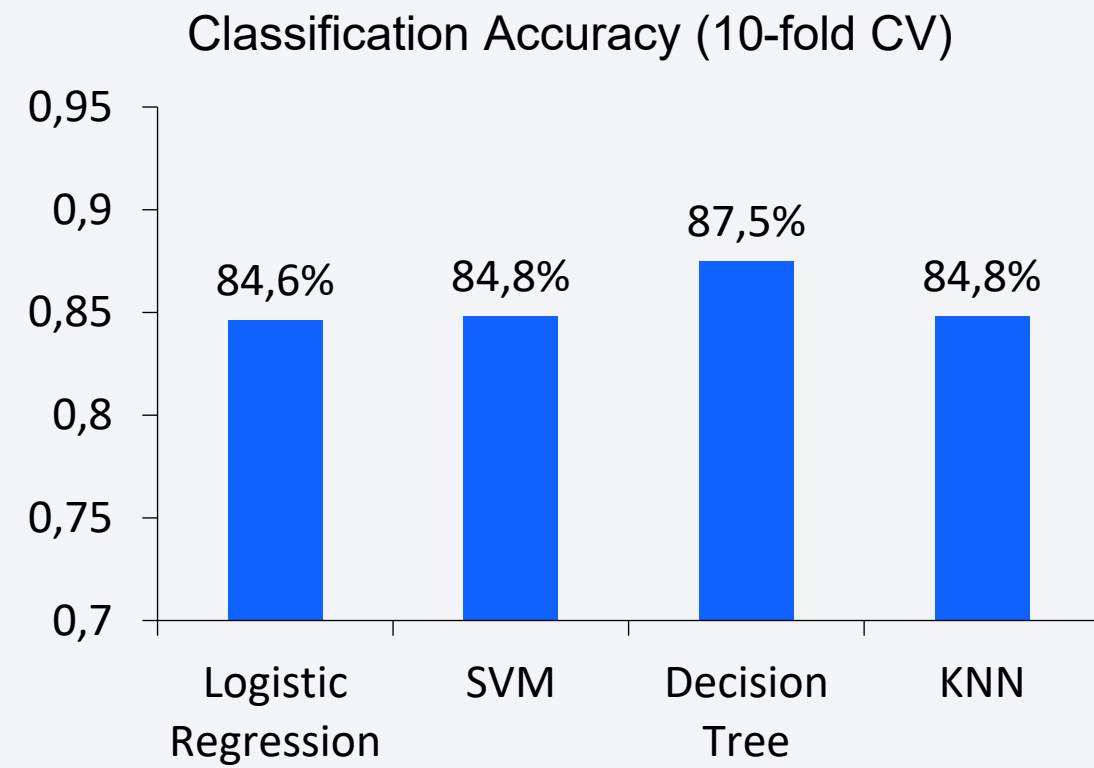
- The scatter plot (all sites, full payload range) plots landing outcome (class) against Payload Mass, colored by Booster Version Category
- Newer booster versions (FT, B4, B5) account for almost all successful (class=1) launches, while the earliest v1.0/v1.1 boosters cluster heavily among failures (class=0)

The background of the slide is a composite image. The left portion is a solid blue field. The right portion is a photograph of a tunnel, likely a subway or transit tunnel, with curved walls and ceiling. The tunnel is illuminated with blue and white lights, creating a sense of depth and motion. The overall aesthetic is modern and technological.

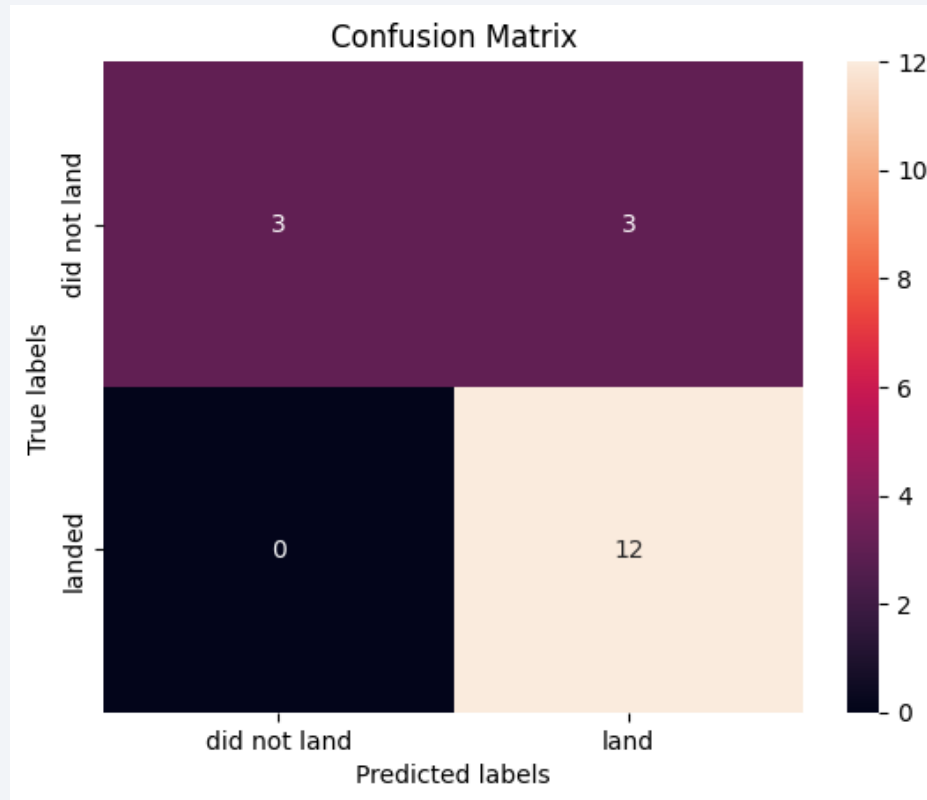
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



Best model: Decision Tree Classifier

Cross-validated accuracy: 87.5%

Test-set accuracy: 83.3% (tied with the other 3 models, due to the small 18-sample test set)

The confusion matrix shows the model correctly identifies most successful landings, with a small number of false positives/negatives typical of a modest dataset size

Conclusions

- Landing success rate has improved substantially over time as SpaceX refined its Falcon 9 booster (from earlier v1.0/v1.1 versions to the modern Block 5)
- Launch site, orbit type and payload mass all show a measurable relationship with landing outcome, with some orbits and payload ranges achieving much higher success rates than others
- SQL and visual EDA were consistent with each other, giving confidence in the cleaned dataset and the patterns identified
- The Decision Tree Classifier was the best-performing model (87.5% cross-validated accuracy), outperforming Logistic Regression, SVM and KNN, and is recommended for predicting Falcon 9 landing outcomes

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!



GRADING SUMMARY

Overall Correctness Score: **100%**

Feedback Breakdown:

[+] Correct: 1

[~] Partially Correct: 0

[X] Needs Improvement: 0

Detailed Feedback Summary:

How to Use This Feedback:

- * Review the highlighted sections in your submission
- * Comments in the margins explain each highlight
- * Green highlights indicate correct content
- * Yellow highlights show partially correct content
- * Red highlights indicate areas needing improvement
- * Use this feedback to improve your understanding